

Brandon T. Pae

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SUMMARY

Research Interests: LLM evaluation, reasoning faithfulness, and safety-critical workflows

Experience: Designed and evaluated LLM systems for document redaction and cultural reasoning, emphasizing controlled experiments, model behavior analysis, and real-world constraints

EDUCATION

Columbia University - GPA: 3.98

New York, NY | May 2026

Bachelor of Science in Computer Science, Minor in Entrepreneurship

- Relevant Coursework - Operating Systems, Advanced C Programming, VR/AR (Unity), Artificial Intelligence, Natural Language Processing, Software as a Service, Parallel Optimization for Robotics

RESEARCH EXPERIENCE

Dexter, *LLM Faithfulness & Document Redaction*

Sep 2025 – Present

- Fine-tuned and evaluated two 8B-parameter LLMs on 20k synthetic documents, measuring a 0% → 5% increase in unfaithful chain-of-thought behavior via controlled evidence-citation tasks and identifying accuracy trade-offs for sensitive document workflows

TinyMPC (A2R Lab), *Model Predictive Control for Resource-Constrained Drones*

Sep 2024 – May 2025

- Modified and evaluated memory-constrained model predictive control algorithms for Crazyflie drones; optimized C/C++ firmware under strict onboard compute limits and empirically validated performance in real-world flight tests

Crackd AI, *Evaluating LLM Humor & Cultural Reasoning*

Sep 2025 – Present

- Built evaluation tests and infrastructure to analyze LLM understanding of humor and cultural context, identifying systematic failure modes in subjective reasoning tasks

WORK EXPERIENCE

Boston Consulting Group (BCG), *Summer Associate*

Boston, MA | Jun 2025 – Aug 2025

- Scoped an AI automation opportunity for an industrial goods client under ambiguous, high-stakes constraints; translated technical feasibility into executive decision-making

FluidityIQ, *Software Engineering Intern*

New York, NY | Jun 2024 – Sep 2024

- Built and deployed a vectorizer for patent-claim retrieval, reducing search API latency by 6x; implemented and evaluated LLM features (summarization, model swapping, multi-user chat), emphasizing reliability in production

Arrow Electronics, *Machine Learning Intern*

Cambridge, MA | Aug 2021 – Aug 2022

- Designed and ran 200+ ML experiments (Optuna, ClearML) to forecast electronics lifecycle metrics; improved the MAPE by 68% relative to baseline models

Capco, *Software Engineering Intern*

New York, NY | Jun 2023 – Aug 2023

- Debugged and validated entitlement and data-access APIs for a U.S. bank, prioritizing failure prevention

LEADERSHIP & COMMUNITY

- **Claude Builder Ambassador (2025–Present):** Co-founded a hands-on LLM builder community in collaboration with Anthropic; organized technical workshops on prompt design, evaluation, and model experimentation
- **Columbia Teaching Assistant (2025–Present):** Mentored 300+ students on web development and debugging (COMS 4170W), emphasizing systematic and controlled iteration
- **Almaworks Director (2022–2024):** Mentored AI startups in the Almaworks Accelerator, across security, education, and robotics, focusing on evaluation criteria and deployment risks

SKILLS

Programming: Python, C/C++, Java, SQL, JavaScript

ML & Evaluation: PyTorch, Keras, Optuna, ClearML

Systems & Infra: LangChain, Docker, AWS, Azure, PostgreSQL, MongoDB